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BITS Pilani
Pilani Campus

CE F230

Civil Engineering Materials

Durability of Concrete



Durability Concern

- ❑ **Maintenance and repair** > \$2 trillion for US and Asia
- ❑ It is estimated that in industrially developed countries over 40% of the total resources of the construction industry are applied to **repair and maintenance** of existing structures, and less than 60% to new constructions.
- ❑ In US and Japan, **outlay for repair** > new construction



Durability of Concrete

- ❑ Concrete should be capable of withstanding the conditions for which it has been designed throughout the life of a structure.
- ❑ Lack of durability can be caused by **external agents** arising from the environment or by **internal agents** within the concrete.
- ❑ Causes can be categorized as **physical, mechanical, and chemical**

Typical Life Span of Concrete Structure



Durability of Concrete

- ✓ **Physical causes**

The action of frost, Thermal movement, Fire

- ✓ **Mechanical causes**

Impact; Abrasion; Erosion

- ✓ **Chemical causes**

Sulfate attack; Sea water attack; Acid attack; Alkali-silica reaction; Corrosion of reinforcement

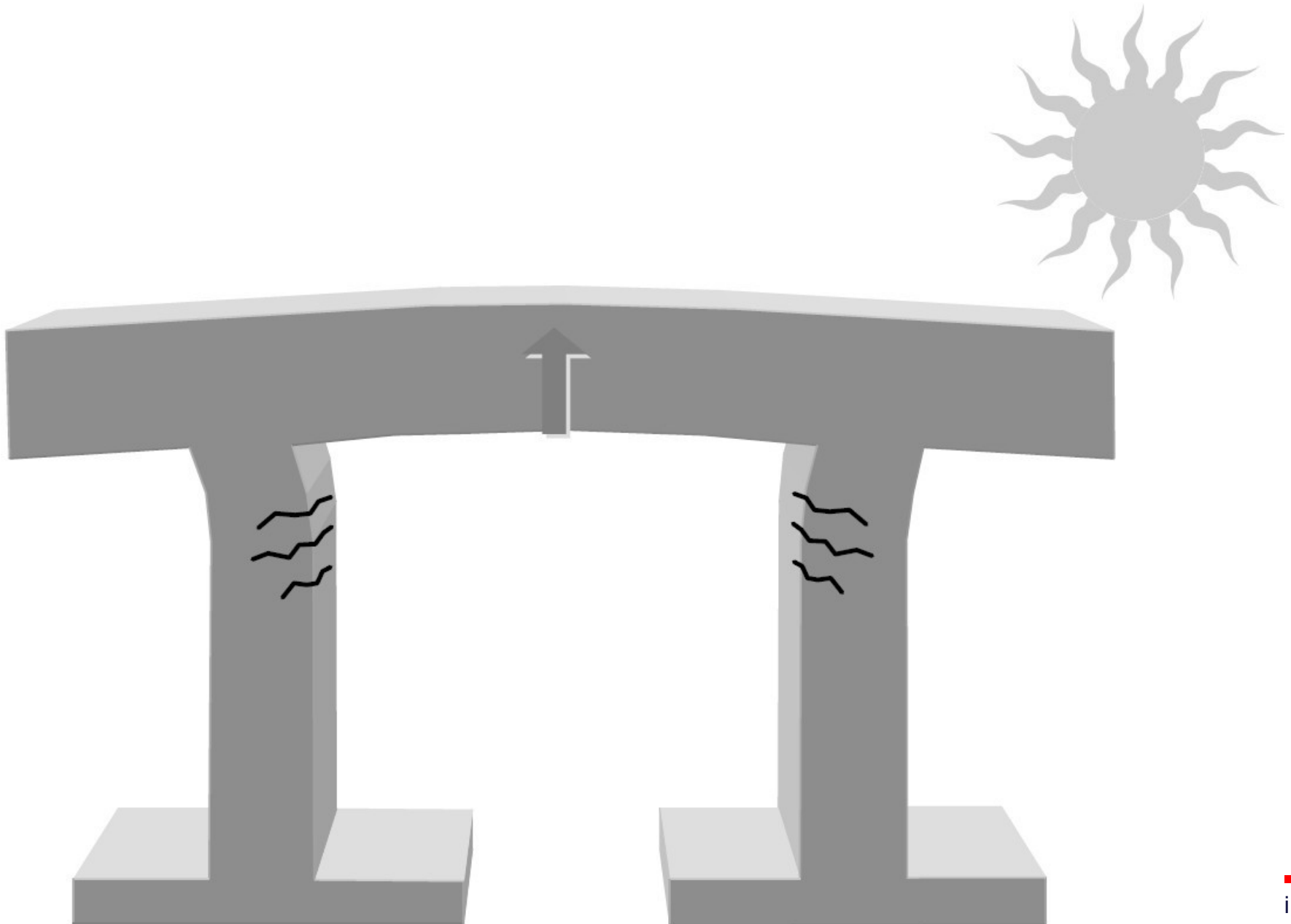
Frost Damage of Concrete

- Concrete can be damaged during freezing and thawing cycles
- Air-entrained concrete have much improved frost resistance



Wang, K., et al. Investigation into Freezing-Thawing Durability of Low Permeability Concrete with and without Air Entraining Agent, IDOT report, 2009

Uneven Thermal Loads



Fire

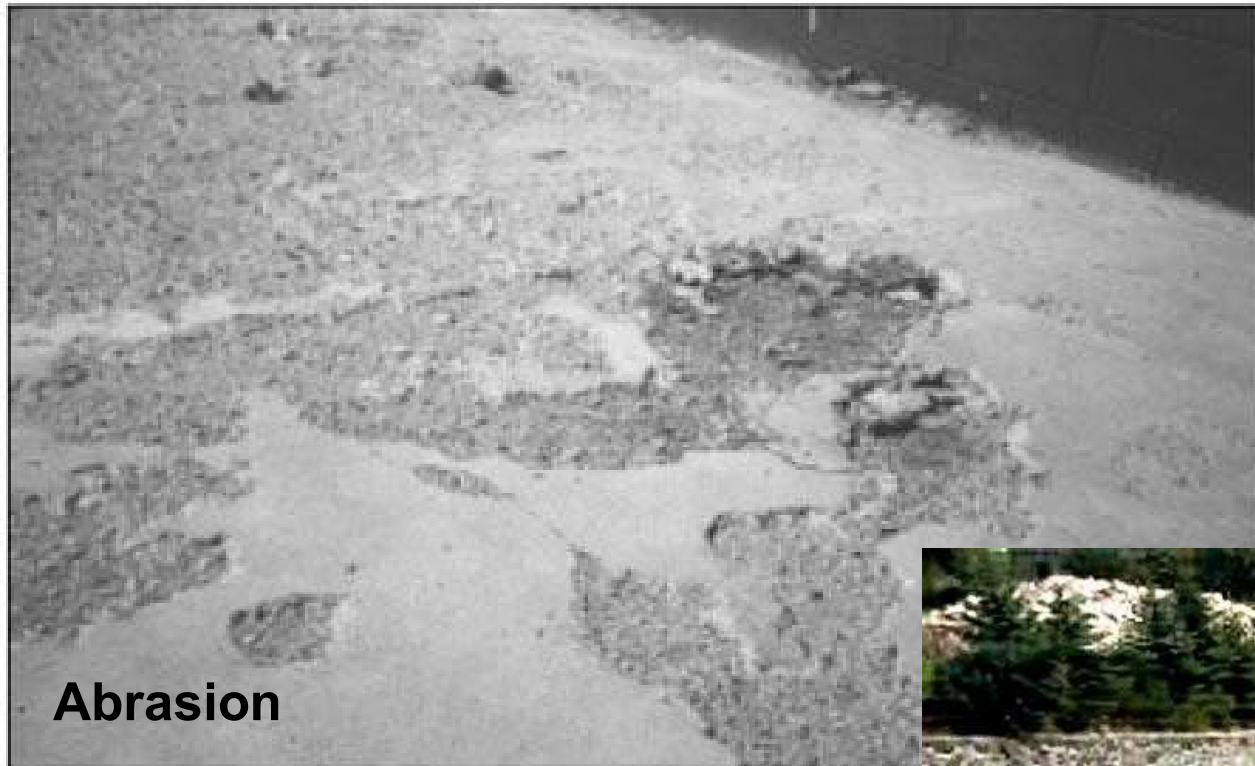


Structural collapse of Windsor tower in Madrid, Spain in 2005

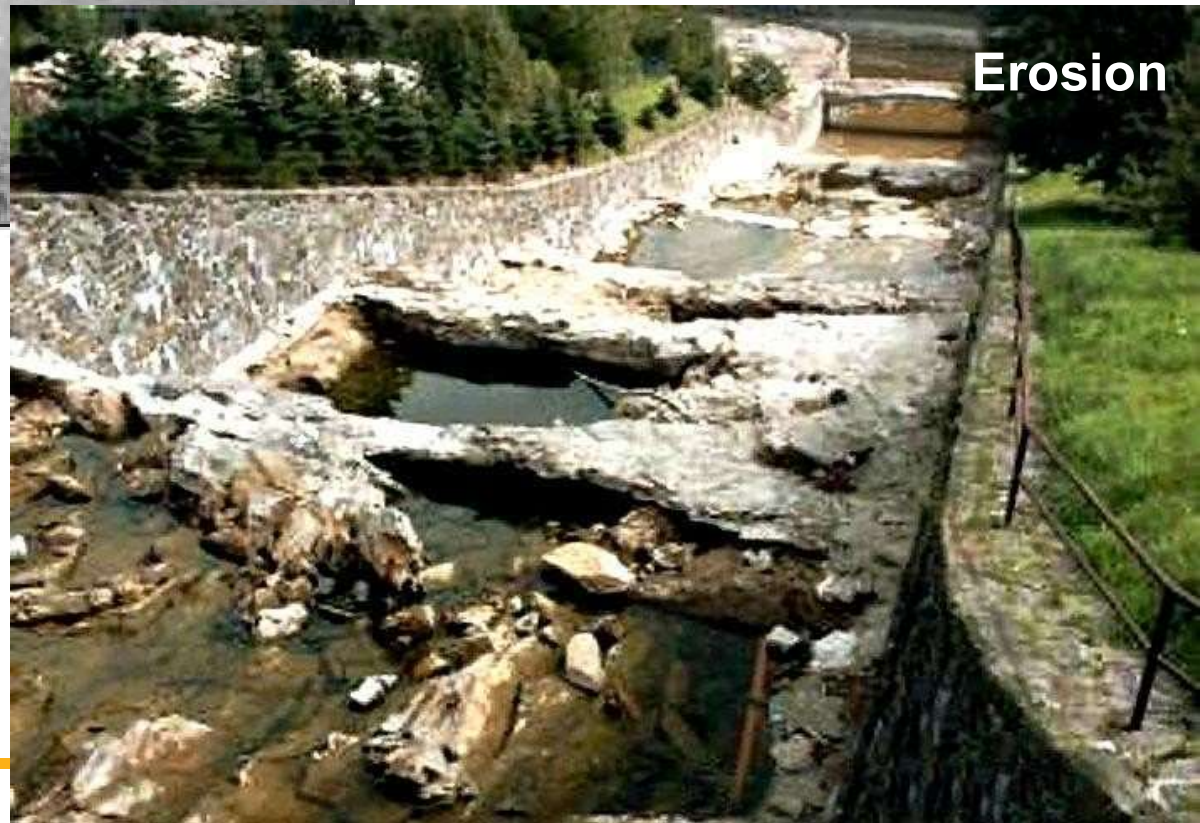


Mont Blanc tunnel fire

Abrasion and Erosion



Abrasion



Erosion

Durability and Transport Properties of Concrete



- ❑ Since the chemical attack takes place within the concrete mass, the attacking agent must be able to **penetrate** through the concrete.
- ❑ Rate at which attacking agent penetrates into concrete depends on its **transport properties**
- ❑ Thus, transport properties of concrete are much related to its **durability**

Transport Properties of Concrete



- ❖ Transport properties are the ease with which liquids or gases can travel through concrete
- ❖ As far as the ease to movement of fluids through concrete is concerned, three transport properties should be distinguished
 - ❖ **Permeability** refers to flow under a pressure differential
 - ❖ **Diffusion** is the process in which a fluid moves under a differential in concentration
 - ❖ **Absorption** is the process in which a liquid moves due to capillary suction

Transport Properties of Concrete



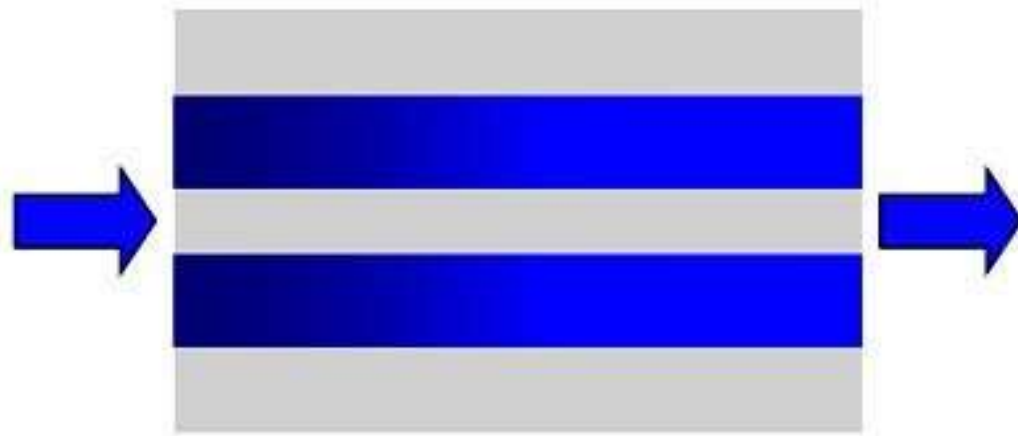
Transport Properties	Mechanism	Example
Absorption	Capillary suction Liquids	Cyclic wetting and drying Marine structure
Permeability	Pressure differential Liquids and gases	Water-retaining structures
Diffusion	Concentration differential Liquids, gases and ions	Sulfate bearing ground waters Foundation elements

Influence of Pore Structure on Concrete Transport Properties



- Transport properties are dependent on
 - Capillary porosity
 - Pore size
 - Continuity of pore system
 - Tortuosity
- Transport properties vs W/C ratio
- Transport properties vs age

Effect of Porosity



Higher Porosity

**Lower Resistance
to Flow ($\text{High } K_p$)**



Lower Porosity

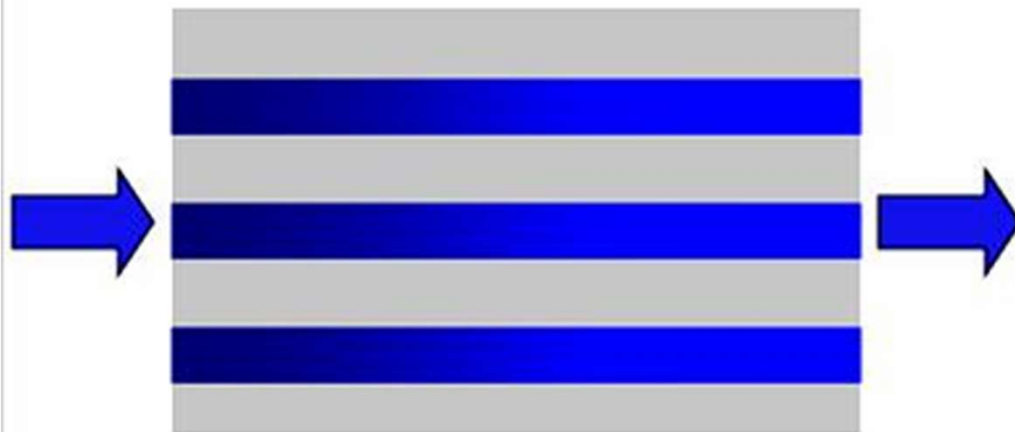
**Greater Resistance
to Flow ($\text{Low } K_p$)**

Effect of Pore Size



One Large Pore

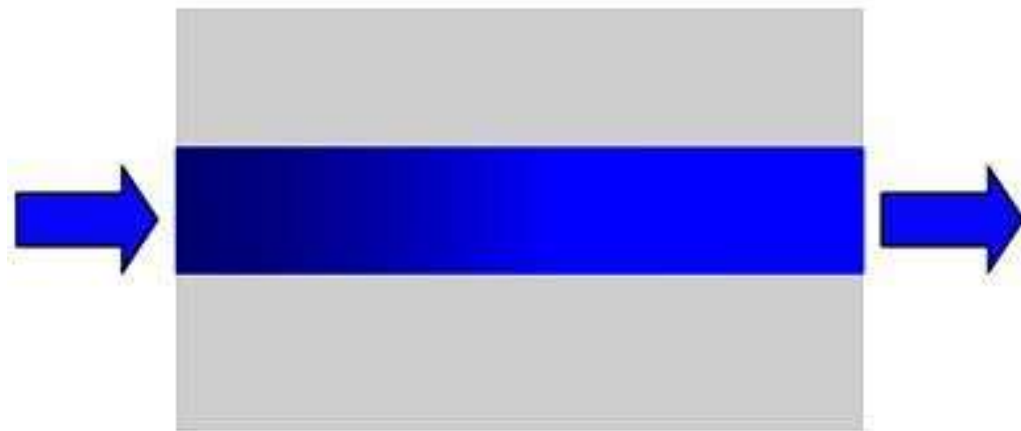
**Lower Resistance
to Flow**



Many Small Pores

**Greater Resistance
to Flow (Low K_p)**

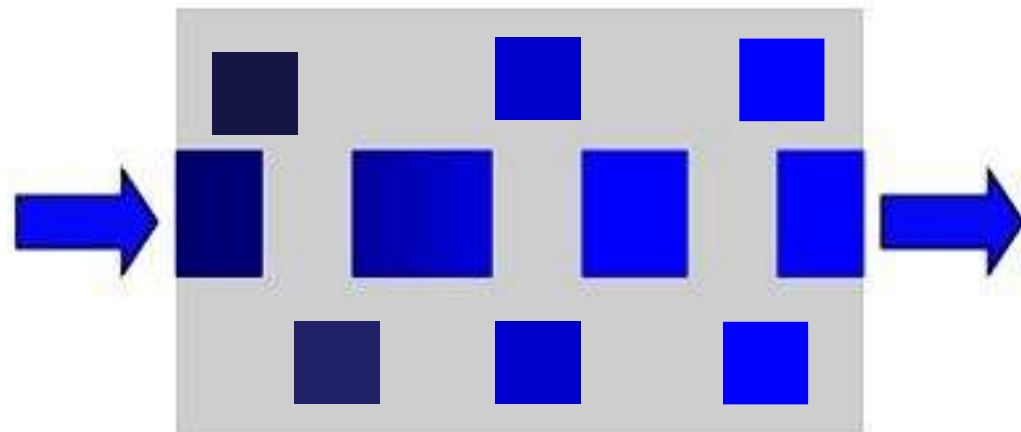
Effect of Connectivity



Connected Pores

**Lower Resistance
to Flow (K_p)**

Same porosity



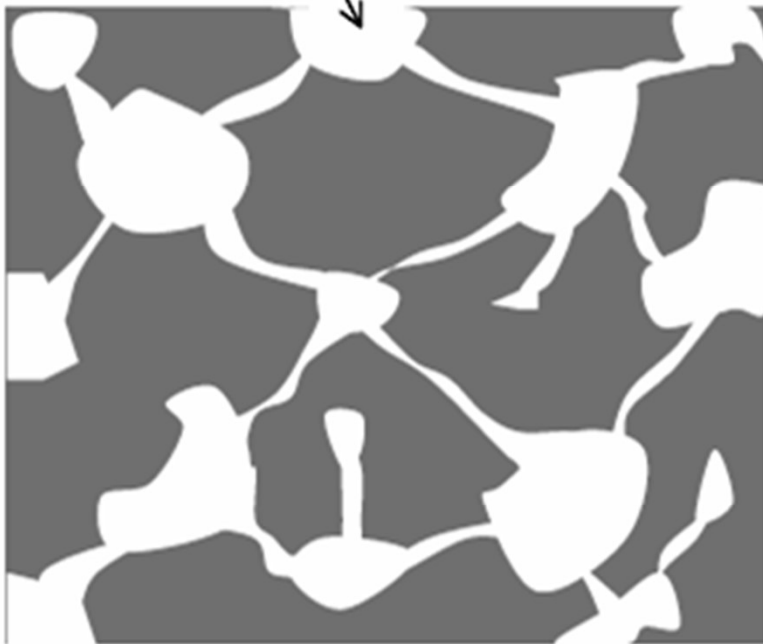
Discrete Pores

**Greater Resistance
to Flow (K_p)**

Schematic Representation of Materials of Similar Porosity



Capillary pores



high permeability-capillary pores interconnected by large passages



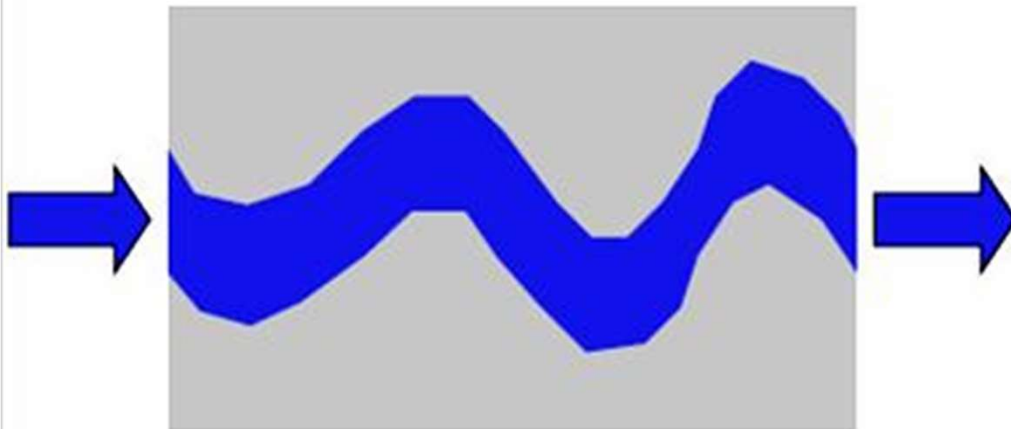
low permeability-capillary pores segmented and only partly connected

Effect of Tortuosity



Straight Pore

**Lower Resistance
to Flow ($\text{High } K_p$)**



Crooked Pores

**Greater Resistance
to Flow ($\text{Low } K_p$)**



Coefficient of Permeability

- Flow in capillary pores in saturated concrete follows Darcy's law for laminar flow through a porous medium:

$$\frac{dq}{dt} \cdot \frac{1}{A} = K \frac{\Delta h}{L}$$

where dq/dt = rate of flow of fluid in m^3/s ,

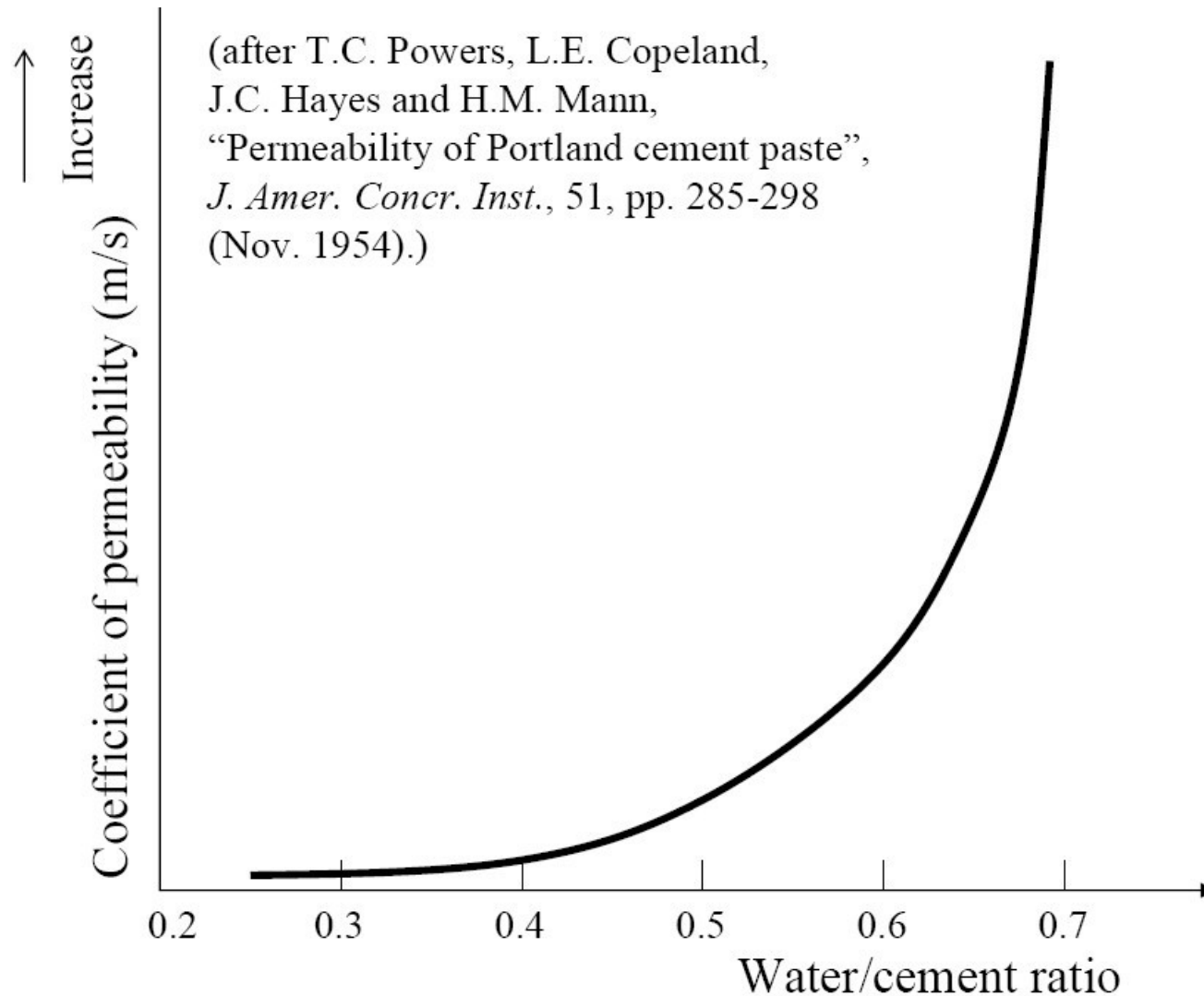
A = cross-sectional area of the sample in m^2 ,

Δh = drop in hydraulic head through the sample in m ,

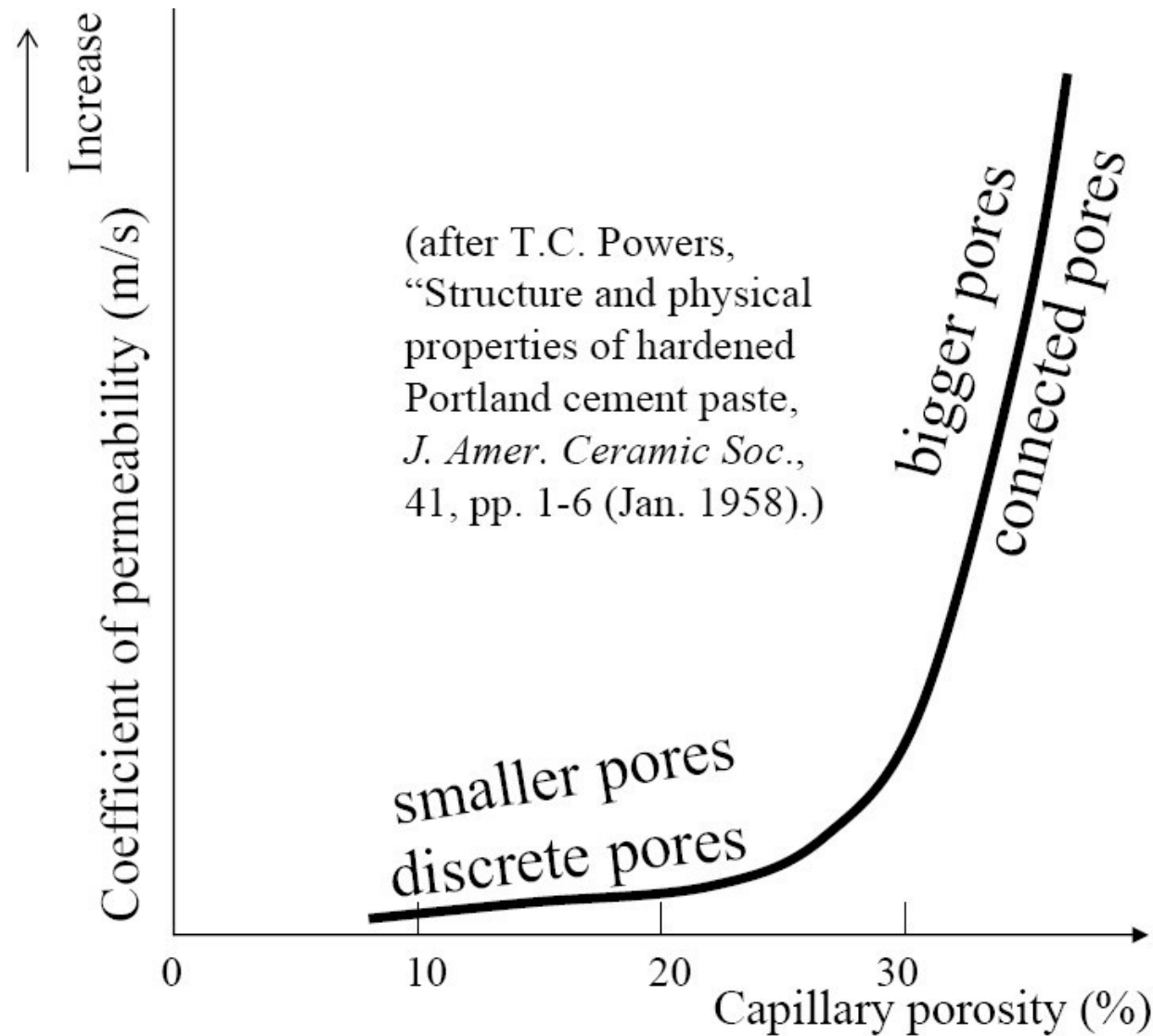
L = thickness of the sample in m ,

K = the coefficient permeability of concrete in m/s ,
dependently of fluid involved (the fluid involved
is generally water).

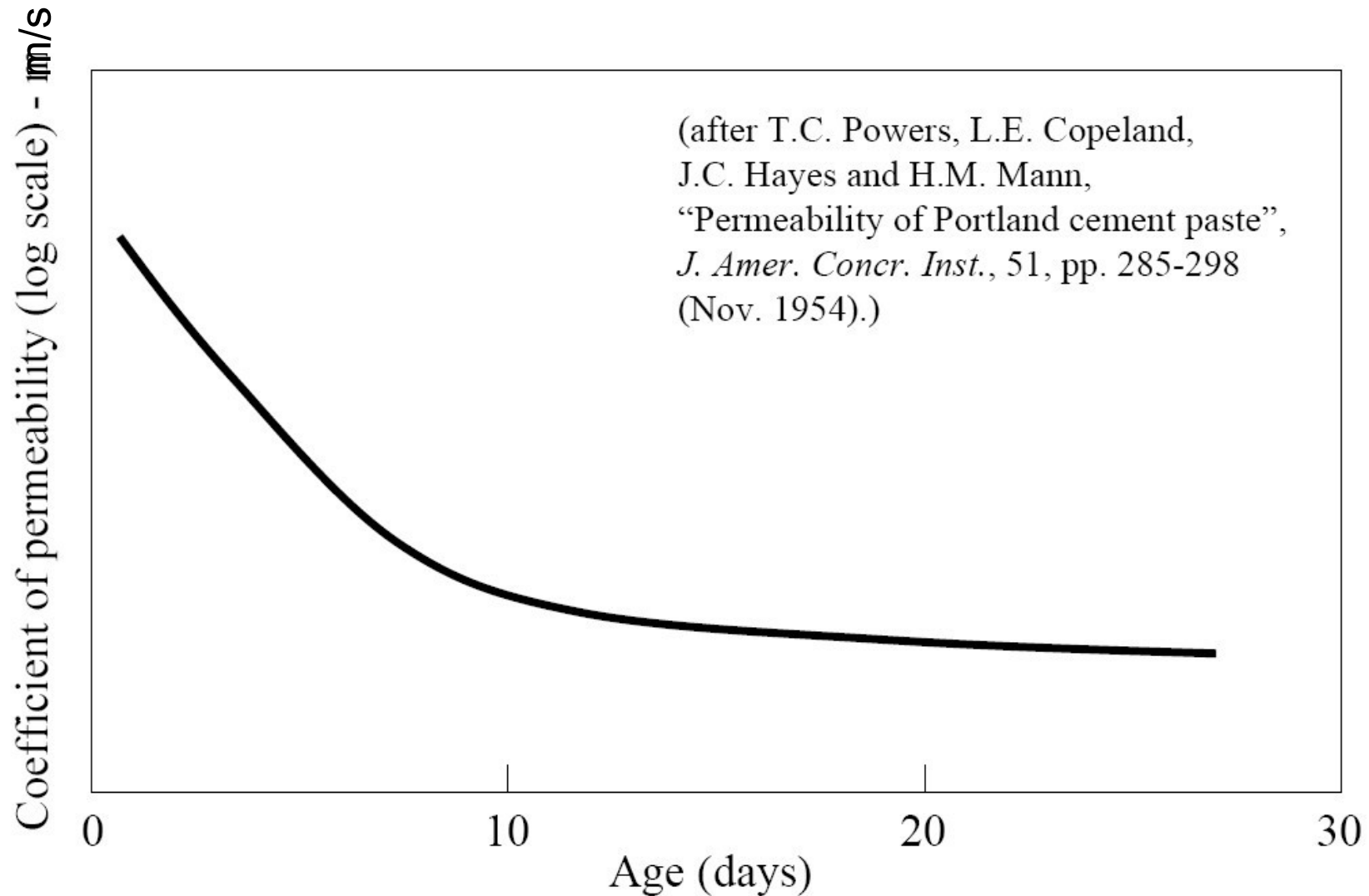
Water Permeability and W/C Ratio for Mature Cement Paste (93% of Cement Hydrated)



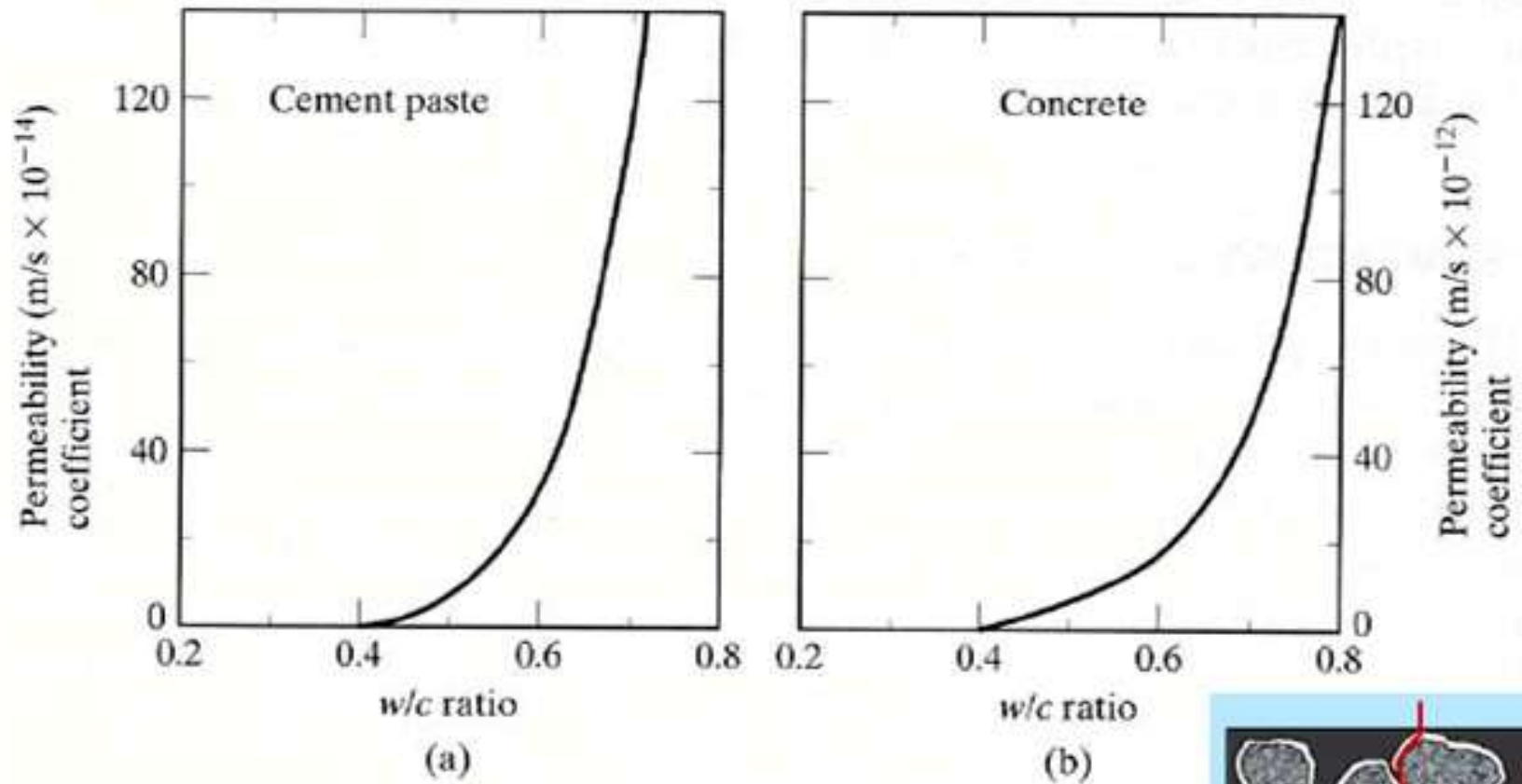
Water Permeability and Capillary Porosity of Cement Paste



Water Permeability of Cement Paste with Progress of Hydration



Water Permeability of Cement Paste & Concrete



(after T.C. Powers, L.E. Copeland, J.C. Hayes and H.M. Mann, "Permeability of Portland cement paste", *J. Amer. Concr. Inst.*, 51, pp. 285-298 (Nov. 1954).)



Chemical Attack

- ❖ Sulfate attack
- ❖ Alkali-silica reaction
- ❖ Corrosion of reinforcement
- ❖ Carbonation

Sulfate Attack

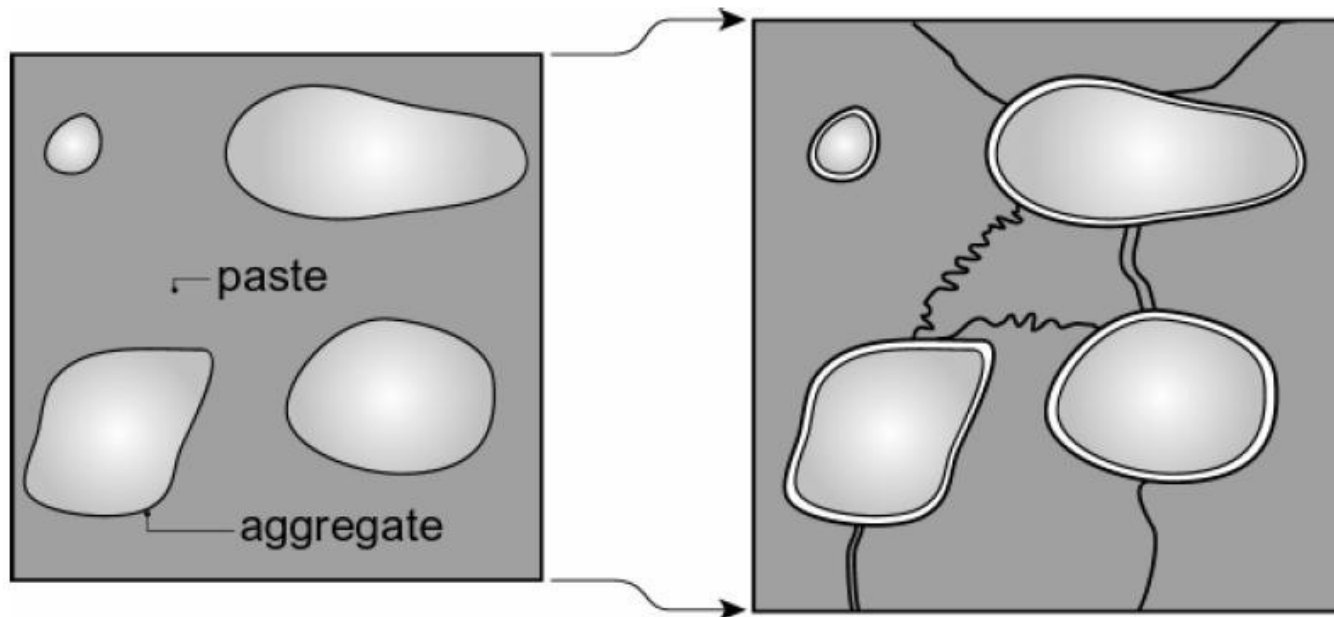
- ❑ Sulfates of sodium, potassium, magnesium, and calcium which occur in soil or in groundwater can react with hydrated cement paste
- ❑ Sulfates in groundwater are usually of natural origin but can also come from fertilizers or from industrial effluents. These sometimes contain ammonium sulfate, which attacks hydrated cement paste.

Sulfate Attack

Mechanism and Process of Deterioration

- Penetration of sulfate ions in concrete
- They **react with calcium hydroxide** (CH) to form gypsum (CSH_2)
- Gypsum further **reacts with C_3A** to form ettringite ($\text{C}_6\text{AS}_3\text{H}_{32}$)
- Formation of ettringite introduces **volume expansion** of hardened concrete which causes internal stresses and cracking

Sulfate Attack



Calcium hydroxide and alumina-bearing phases of hydrated portland cement are more vulnerable to attack by sulfate ions.

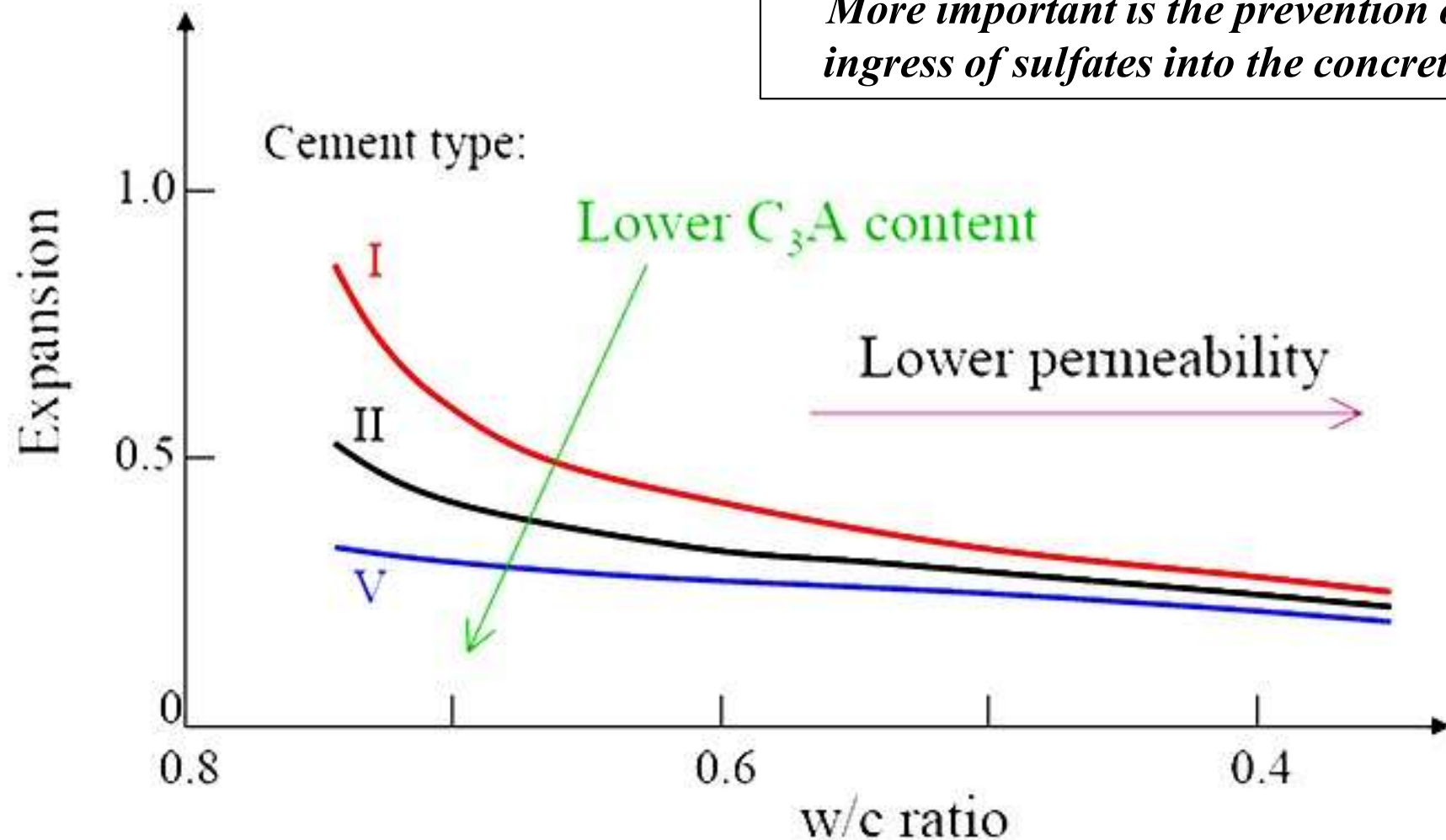
Sequence of Sulfate Attack

Reaction Step	Chemical Reaction	Relevant Factor in Concrete	Control Strategy
1	SO_4^{2-} (external) → SO_4^{2-} (internal)		
2	$\text{Ca}(\text{OH})_2 + \text{SO}_4^{2-}$ → gypsum + 2OH^-		
3	C_3A or $\text{C}_4\text{A}\text{SH}_{12}$ + gypsum → ettringite		

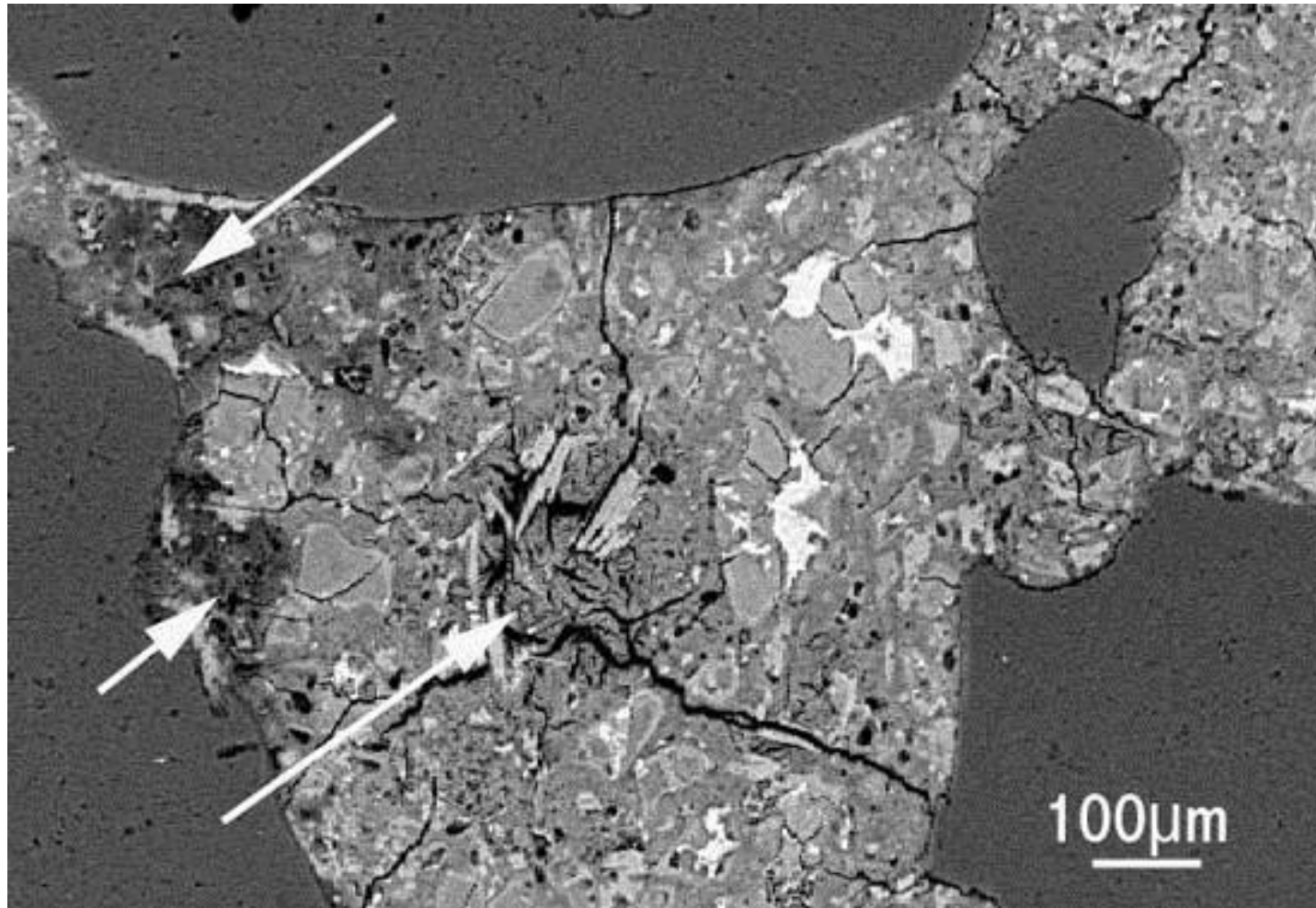
Effects of Cement Type and W/C Ratio on Sulfate Attack



More important is the prevention of ingress of sulfates into the concrete.



Sulfate Attack



Source: N. Winter, Understanding Cement



Alkali-Silica Reaction (ASR)

- Concrete can be damaged by a chemical reaction between the active silica constituents of the aggregate and the alkalis in the cement; this process is known as alkali-silica reaction (ASR)
- Reactants of ASR: **reactive silica, alkalis, and water**
- Not all types of aggregates contain reactive forms of silica

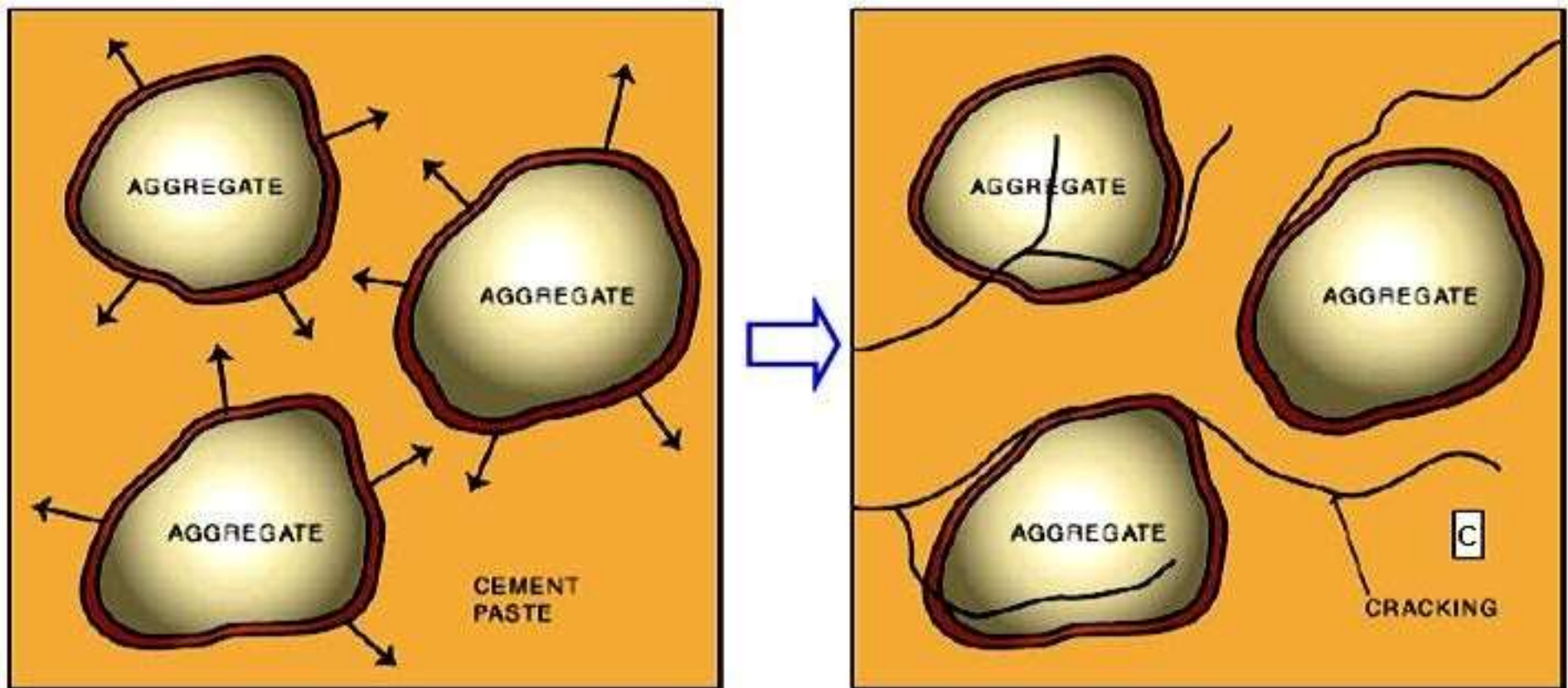
Mechanism and Process of Deterioration of ASR



- The reaction starts with the attack on the siliceous minerals in the aggregate by the alkaline hydroxides derived from the alkalis in the cement
- The ***alkali-silicate gel*** formed attracts water by absorption or by osmosis and thus **tends to increase in volume**
- Since the gel is confined by the surrounding cement paste, internal pressures result and eventually lead to **expansion, cracking and disruption** of the cement paste (pop-outs and spalling) and to **map cracking** of the concrete

Expansion of Alkali-Silica Gel

When the expansionary pressure exceeds the tensile strength of the concrete, the concrete cracks.



Concrete Failure Due to ASR



- Map cracking or pattern cracking
- The crack width can range from 0.1 mm to as much as 10 mm in extreme cases
- The cracks are rarely more than 25 mm, or at most 50 mm, deep



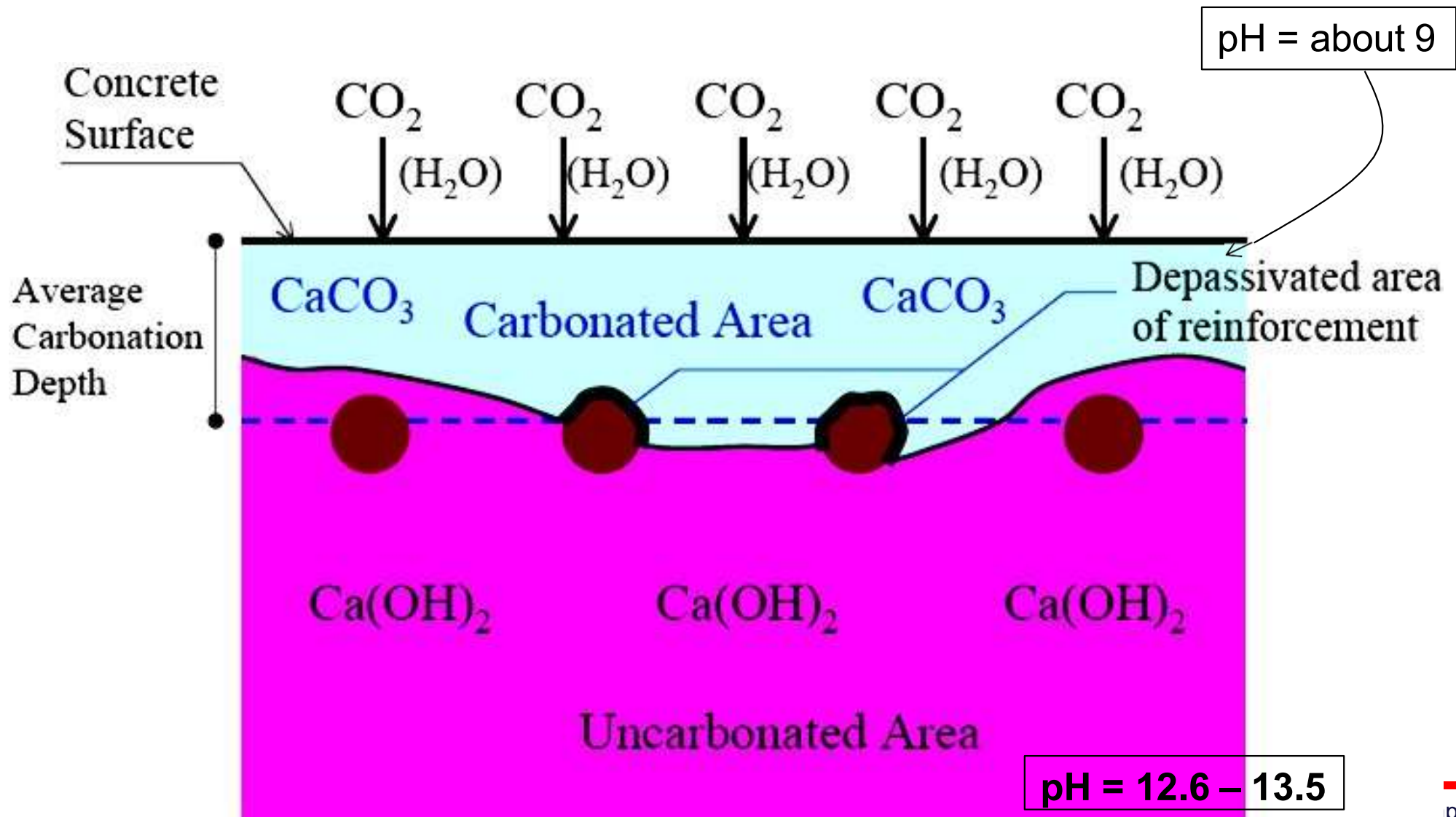
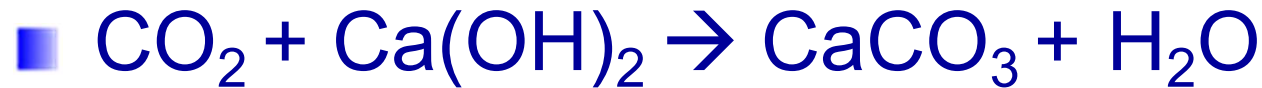
Source: AASHTO Innovative Highway Technologies



Control Strategy for ASR

- ✓ Do not use alkali-reactive aggregate
- ✓ Do not use cements having a high alkali content
- ✓ Use suitable proportions of either fly ash or slag in the concrete
- ✓ Ensure the environment is dry

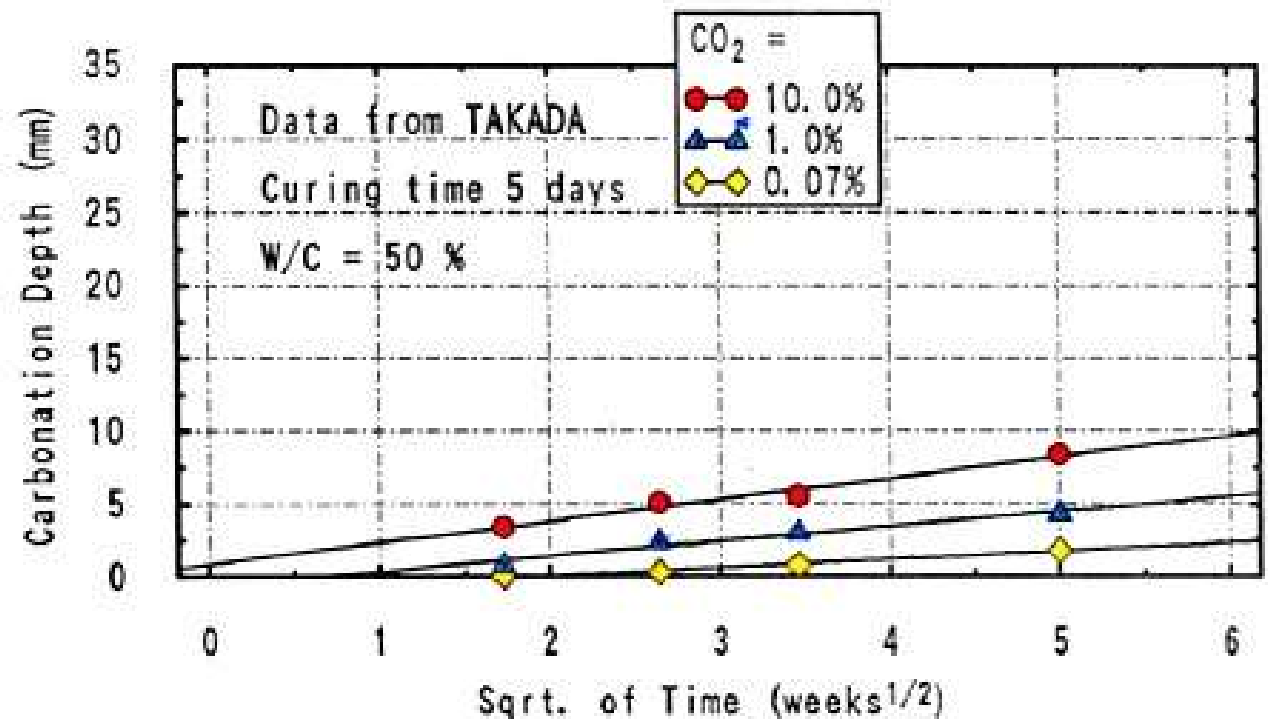
Carbonation



Depth of Carbonation

$$X = A\sqrt{t}$$

- X = carbonation depth
- t = exposure time
- A = carbonation rate which is dependent on many factors.



Factors Affecting Rate of Carbonation



Factors	Mechanisms	Results
W/C ratio		
Initial water curing period		
CO ₂ concentration		
Temperature		



Corrosion of Reinforcement

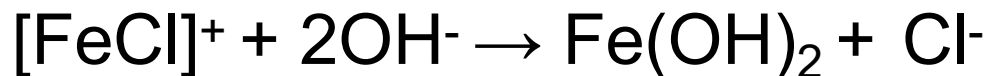
- Structural concrete is usually designed in combination with reinforcing steel bars due to the low tensile strength of the concrete
- The strongly alkaline nature of $\text{Ca}(\text{OH})_2$ (pH of about 13) prevents the corrosion of the steel reinforcement by the formation of a thin protective film of iron oxide on the metal surface; this protection is known as passivity
- As long as this film is stable, the steel is immune to corrosion.

Two Main Causes Lead to Disruption of Passivation Film



■ Penetration of chlorides into the concrete

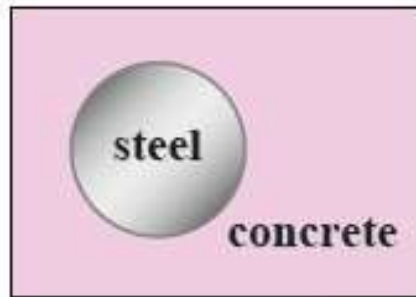
- Depassivation occurs when the chloride ion content reaches a value in the range of 0.2% to 0.4% by weight of the Portland cement in the concrete adjacent to the steel



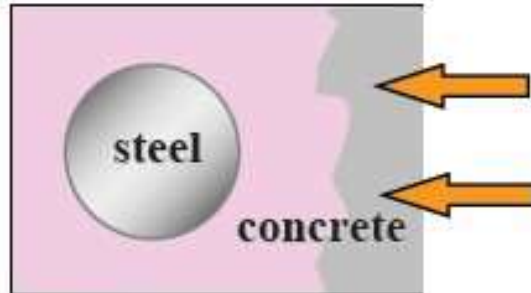
■ Carbonation of the concrete leading to reduction in pH

- Depassivation occurs as pH approaches 9

Carbonation to Corrosion



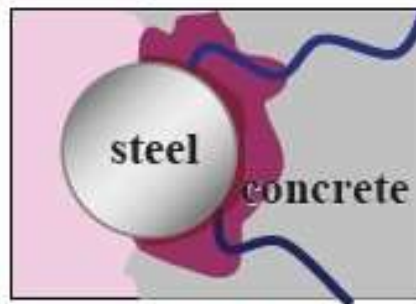
Good quality concrete (pH=13) steel is passivated.



Carbon dioxide enters, pH begins to drop. Steel is not yet affected.

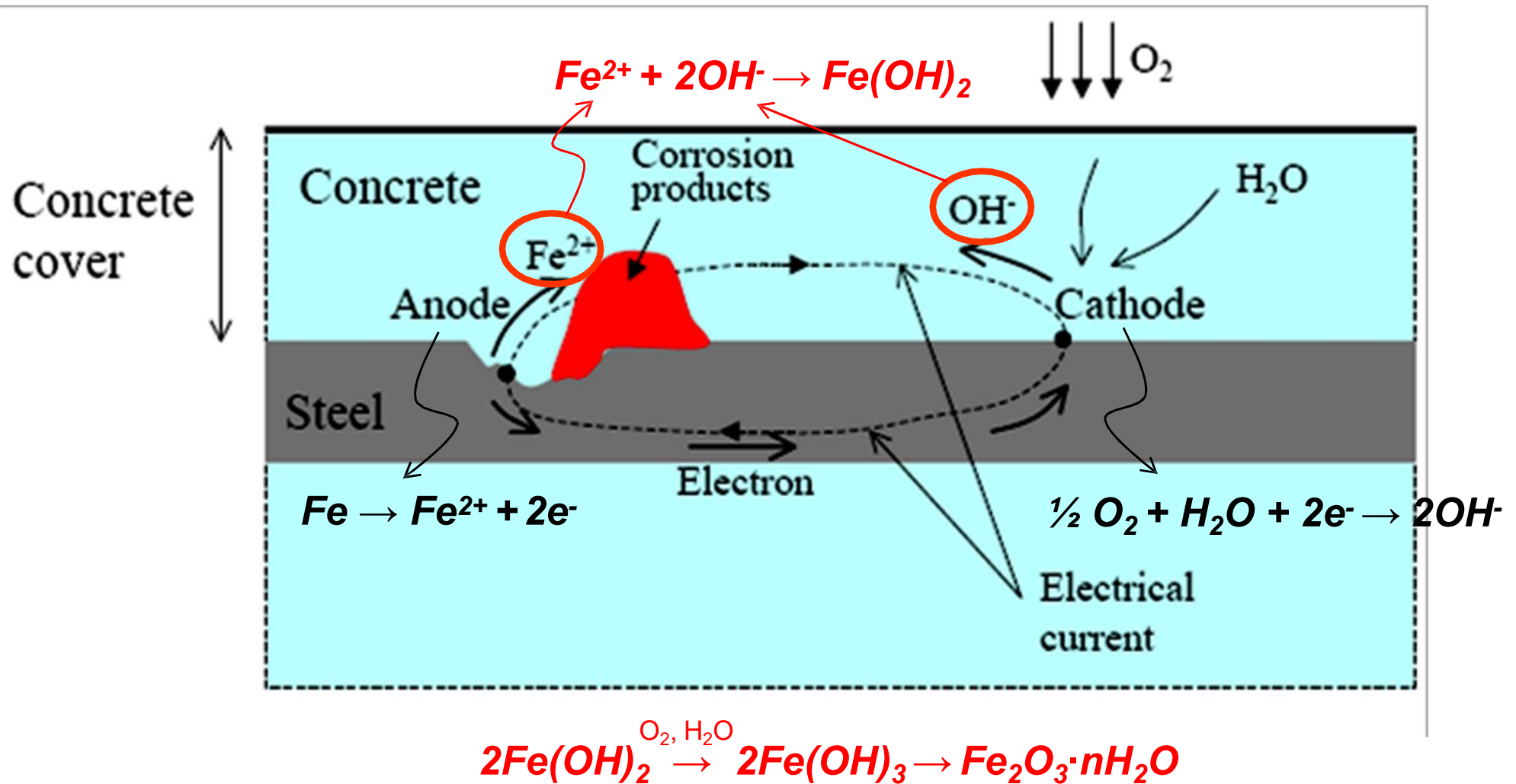


pH at steel drops below 9, corrosion begins.

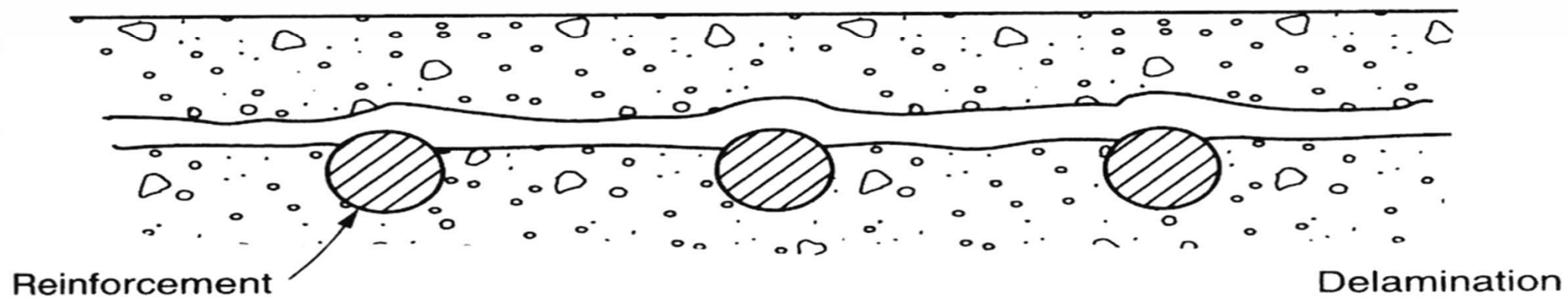
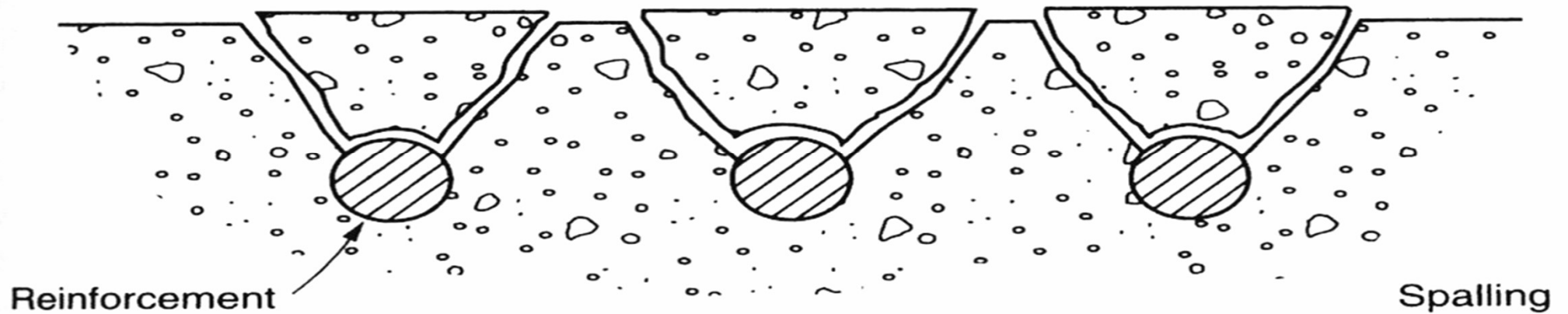
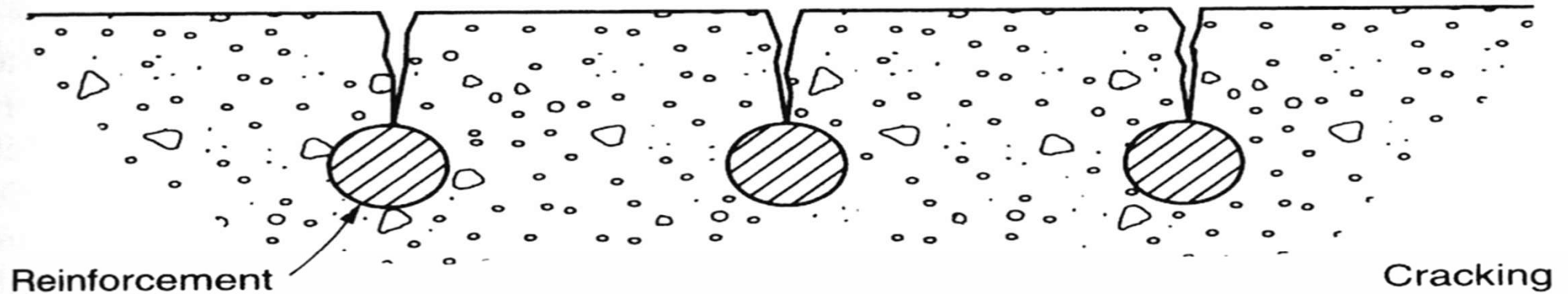


Volume expansion of rust causes cracking and spalling.

Process of Corrosion of Steel Reinforcement in Concrete



Damage Induced by Corrosion



Corrosion Introduced Cracking and Spalling



Corrosion Protection

Prevention of corrosion lies in controlling the ingress of chlorides by the thickness of cover to reinforcement and by the penetrability of the concrete in the cover

The five C's

- Proper **C**over of at least 50 mm
- Lower W/C ratio and sufficient **C**ement
- Sufficient **C**uring time
- Denser concrete, good **C**ompaction
- Avoid using **C**hlorides
- Pozzolan addition, slag addition